# CAO-DF - Entity-Relationship Diagram (datasets and result tables)

**WORKLOAD**
(workload_index.csv / validation_index.csv)

- ● workload_id : text «PK»

data_volume : text
file_count_level : text
shuffle_volume_level : text
seed : int
num_orders : int
actual_file_count : int
actual_size_mb : float
orders_path : text
customers_path : text

described by

**MANIFEST**
(manifest.json)

- ● workload_id : text «PK,FK»

seed : int
num_orders_generated : int
num_customers_generated : int
actual_order_file_count : int
actual_order_size_bytes : int
actual_order_size_mb : float
requested_shuffle_fraction : float
actual_positive_amount_fraction : float

contains

**CUSTOMER**
(Parquet)

- ● customer_id : long «PK»

region : north | south | east | west
segment : retail | enterprise | smb

places

**ORDER**
(Parquet)

- ● order_id : long «PK»

customer_id : long «FK»
product_category : text
amount : double
order_date : text (date)

contains

measured in

**EXPERIMENT_RUN**
(experiment_results.csv)

- ● run_id : text «PK»

workload_id : text «FK»
experiment_action : A1 | A2 | VAL
config_label : text
shuffle_partitions : int
adaptive_enabled : bool
repetition_number : int
is_warmup : bool
wall_clock_seconds : float
executor_run_time_ms : long
executor_cpu_time_ms : long
shuffle_bytes_written : long
disk_bytes_spilled : long
peak_memory_percent : float
avg_cpu_percent : float
... (34 columns in total)

median per cell

**COST_CELL**
(cost_per_cell.csv)

- ● experiment_action : A1 | A2 «PK»
- ● workload_id : text «PK,FK»
- ● config_label : text «PK»

shuffle_partitions : int
n : int
median_wall_s : float
median_executor_run_s : float
median_cost_P1, iqr_cost_P1 : float
median_cost_P2, iqr_cost_P2 : float

combined into options

**BREAKEVEN_OPTION**
(breakeven_table.csv)

- ● workload_id : text «PK,FK»
- ● action : A1_search | A1_rule | A2 «PK»
- ● proxy : P1 | P2 «PK»

chosen_setting, search_settings : text
crun_static_usd, crun_action_usd : float
copt_usd, saving_per_run_usd : float
saving_pct : float
static_iqr_usd : float
saving_exceeds_static_iqr : bool
n_star : float
tc_static_N{1,5,10,50} : float
tc_action_N{1,5,10,50} : float
worthwhile_N{1,5,10,50} : bool

price sensitivity

**COST_SENSITIVITY**
(cost_sensitivity.csv)

- ● price_factor : 0.5 | 1.0 | 1.5 «PK»
- ● workload_id : text «PK,FK»
- ● action : text «PK»
- ● proxy : P1 | P2 «PK»

price_per_vcpu_hour_usd : float
chosen_setting : text
n_star : float
net_saving_N10_usd : float
worthwhile_N10 : bool

validated in

**VALIDATION_DECISION**
(validation_decisions.csv)

- ● workload_id : text «PK,FK»
- ● proxy : P1 | P2 «PK»
- ● n_runs : 1 | 5 | 10 | 50 «PK»

framework_choice, performance_only_choice,
oracle_choice : Nothing | Tune | Compact
a1_neighbours, a2_neighbours : text
est_saving_tune, est_saving_compact : float
measured_saving_tune, measured_saving_compact : float
tc_static_usd, tc_framework_usd : float
tc_performance_only_usd, tc_oracle_usd : float
tc_tune_rule_usd : float
framework_reduction_pct : float
performance_only_reduction_pct : float
oracle_reduction_pct : float
framework_regret_pct : float
framework_vs_static : win | tie | loss
performance_only_vs_static : win | tie | loss
framework_matches_oracle : bool

summarized into

**VALIDATION_SUMMARY**
(validation_summary.csv)

- ● proxy : P1 | P2 «PK»
- ● strategy : framework | performance_only «PK»

cases, wins, ties, losses : int
mean_reduction_pct : float
ci95_low, ci95_high : float
total_cost_reduction_pct : float
matches_oracle : int
max_regret_pct : float

left out in

**LOO_RESULT**
(framework_loo.csv)

- ● workload_id : text «PK,FK»
- ● action : A1 | A2 «PK»
- ● proxy : P1 | P2 «PK»
- ● n_runs : int «PK»

neighbours : text
predicted_apply : bool
measured_apply : bool
agree : bool

- input data (generated)
- measured runs (raw)
- cost model outputs
- framework validation outputs